04820KA001 Mono Canada II

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Comment: Modified to Connect 2nd Diameter Sens

Sysmac Studio Module Version: 1.60.0.64010

1.PLC

1-7.POUs

1-7-1.Programs

1-7-1-1.TaglMonoTRM

1-7-1-1-1.Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Comment
VAR						
Diff	REAL			False	False	Minimum or maximum speed difference
DiffMin	REAL			False	False	Minimum speed difference
DiffMax	REAL			False	False	Maximum speed difference
Clock100ms	BOOL			False	False	100 ms clock
MemPos	DINT			False	False	Previous encoder count
DeltaPulse	DINT			False	False	Encoder count difference
Vel_mms	REAL			False	False	Speed in mm/s
Allarme	ARRAY[0..32] OF BOOL			False	False	Alarm
mm	REAL			False	False	Current millimetres
T_PlcOn	TON			False	False	PLC active timeout
AuxOk	BOOL			False	False	Auxiliaries OK
CmdStartMotore	BOOL			False	False	Motor start command
CntMtOk	BOOL			False	False	Metre count reached
CmdOkMozzoOut	BOOL			False	False	Hub extraction enable command
CmdFreno	BOOL			False	False	Brake command
T_StopVent	TOF			False	False	Fans stop timer
TimStopVen	TIME			False	False	Fans stop timeout
CmdVentole	BOOL			False	False	Fans command
PressInAuto	BOOL			False	False	Rider roll in automatic
PressInCont	BOOL			False	False	Rider roll in continuous
CmdBlokPres	BOOL			False	False	Rider roll lock command
CmdAlzoPres	BOOL			False	False	Rider roll raise command
MemPot1	INT			False	False	Work potentiometer value
Clock10ms	BOOL			False	False	10 ms clock
VelSet_mms	REAL			False	False	Set speed in mm/s
ModVelInv	REAL			False	False	Inverter speed modifier
T_PosPresOff	TON			False	False	Rider roll photocell OFF timer
T_PosPresOn	TON			False	False	Rider roll photocell ON timer
TimPosPresOff	TIME			False	False	Rider roll photocell OFF timeout
TimPosPresOn	TIME			False	False	Rider roll photocell ON timeout
M_DiamBo	BOOL			False	False	Roll diameter photocell intercepted
RappAnalOut	REAL			False	False	Analog output ratio
RappAnalIn	REAL			False	False	Analog input ratio
T_StartLow	TON			False	False	Slow start timer
T_DaimBobOff	TON			False	False	Roll diameter photocell OFF timer
DiamBobOk	BOOL			False	False	Roll diameter reached
B	INT			False	False	
AlarmStrobe	BOOL			False	False	Alarm message display strobe
T_BanAll	TON			False	False	Alarm display timer
RangeOK	BOOL			False	False	Range reached
VelMax_mms	REAL			False	False	Maximum speed in mm/s
DisPuSlow	BOOL			False	False	Inhibit slow button
Diminuisci2	REAL			False	False	Deceleration ramp
Aumenta2	REAL			False	False	Acceleration ramp

	pippo	BOOL			False	False	(scratch/test variable)
	MemPot2	INT			False	False	Final potentiometer value
	mm_Imp	REAL			False	False	mm per encoder pulse
	Rallenta	BOOL			False	False	Slow down
	Arresto	BOOL			False	False	Stop
	T_EncoderFermo	TON			False	False	(?) Encoder stopped timer
	AnalRes	UINT			True	False	Analog board resolution
	Mt_To_mm	REAL			False	False	m/min to mm/s factor
	Mm_To_mt	REAL			False	False	mm/s to m/min factor
	Pluto	UDINT			False	False	(scratch/test variable)
	cippo	WORD			False	False	(scratch/test variable)
	Pot1_Word	WORD			False	False	(?) Pot 1 raw word
	Pot1_WordSW	WORD			False	False	(?) Pot 1 raw word, byte-swapped
	AnalRes2	WORD			False	False	(?) Analog resolution 2
	AnalRes2SW	WORD			False	False	(?) Analog resolution 2, byte-swapped
	Pot2_Word	WORD			False	False	(?) Pot 2 raw word
	Pot2_WordSW	WORD			False	False	(?) Pot 2 raw word, byte-swapped
	InvRef2	UINT			False	False	(?) Inverter reference 2
	InvRefSW	WORD			False	False	(?) Inverter reference, byte-swapped
	InvRefWW	WORD			False	False	(?) Inverter reference, word
	IniRampe	BOOL			True	False	(?) Ramp initialisation
	T_MemNocOk	TON			False	False	(?) Core value stored timer
	VascaOffDiam	BOOL			False	False	(?) Tub/cradle off by diameter
	RallentaDiam	BOOL			False	False	(?) Slow down by diameter
	ArrestoDiam	BOOL			False	False	(?) Stop by diameter
	T_PagIni	TON			False	False	(?) Start page timer
	T_PulStop	TON			False	False	(?) Stop button timer
	RangeMM	REAL			True	False	(?) Sensor range in mm
	FineR	REAL			True	False	Maximum sensor reading (full scale)
	ZM	REAL			True	False	Sensor dead zone
	RangePt	REAL			True	False	(?) Sensor range in raw counts
	FineR_Pt	REAL			True	False	(?) Full scale in raw counts
	ZMpt	REAL			True	False	(?) Dead zone in raw counts
	Ris	REAL			True	False	(?) Result
	Tensione	REAL			True	False	(?) Voltage
	Old	BOOL			False	False	(?) Previous value
	V_InAnal	REAL			True	False	Analog input voltage
	loop	BOOL			False	False	(?) Loop index
VAR_EXTERNAL							
	VelSet	REAL				False	Set speed
	H_Range	BOOL				False	Speed too high
	PercDiff	REAL				False	Speed difference percentage
	EncPos	DINT				False	Encoder counter
	Vel_mtmin	REAL				False	Speed in m/min
	mt	REAL				False	Current metres
	I_RsetCnt	BOOL				False	Metre counter reset button
	L_Range	BOOL				False	Speed too low
	I_AuxOn	BOOL				False	Auxiliaries ON
	I_BarOk	BOOL				False	Light barriers reset
	I_DiamBob	BOOL				False	Roll diameter photocell
	O_Motore	BOOL				False	Motor start
	I_PosPress	BOOL				False	Rider roll position sensor
	I_PresPress	BOOL				False	Rider roll pressure switch

I_PressAuto	BOOL				False	Rider roll in automatic
I_PressCont	BOOL				False	Rider roll in continuous
I_Marcia	BOOL				False	Start button
I_PorteClose	BOOL				False	Side doors closed
I_PastTerm	BOOL				False	Motor thermal switch
I_ThVentOk	BOOL				False	Motor (fan) thermal OK
I_Arresto	BOOL				False	Stop button
I_MozzoInPos	BOOL				False	Hub/chuck in extraction position
I_RipaClose	BOOL				False	Front guard closed
O_Allarme	BOOL				False	Alarm lamp
O_PorteOpen	BOOL				False	Doors open lamp
O_OkScaricatori	BOOL				False	Unloaders enable - opening
I_ProtEsc	BOOL				False	Guards-bypassed selector switch
O_ProtEscluse	BOOL				False	Guards bypassed lamp
O_Freno	BOOL				False	Brake command
TimStopVenP	REAL				False	Fans stop timeout (from panel)
O_Ventole	BOOL				False	Brake and motor fans
O_PressAuCo	BOOL				False	Rider roll auto/continuous lamp
O_PressUp	BOOL				False	Raise rider roll
O_BlockPres	BOOL				False	Lock rider roll
Fix10Vdc	INT				False	Fixed 10 Vdc generator
InvRef	INT				False	Inverter speed reference
EncOn	BOOL				False	Encoder enable
EncRset	BOOL				False	Encoder reset
TimPosPresOffP	REAL				False	Rider roll photocell OFF timeout (panel)
TimPosPresOnP	REAL				False	Rider roll photocell ON timeout (panel)
VelMax	REAL				False	Maximum speed
I_PuSlow	BOOL				False	Slow speed button
O_L_MotOn	BOOL				False	Motor ON lamp
AllarmeOn	BOOL				False	Alarm active
BuzzerOn	BOOL				False	Panel buzzer command
I_InvAll	BOOL				False	Inverter in alarm
Msg	INT				False	Message
Indice	INT				False	Active alarm index
MotStop	BOOL				False	Motor stopped
MotOnH	BOOL				False	Motor running fast
MotOnL	BOOL				False	Motor running slow
Clock1s	BOOL				False	1 s clock
ManMode	BOOL				False	Manual mode
SetVelMan	INT				False	% speed in manual
H_RsetCnt	BOOL				False	Panel counter reset button
I_RipaOpen	BOOL				False	Front guard open
O_OkScarIn	BOOL				False	Unloaders enable - closing
VelSetMan	REAL				False	Speed in manual
O_AccDec2	BOOL				False	Second acc/dec ramp
Aumenta	REAL				False	Acceleration coefficient in run
Diminuisci	REAL				False	Deceleration coefficient in run
DiminuisciH	REAL				False	Acceleration coefficient at start (orig. comment; name suggests decel)
AumentaH	REAL				False	Deceleration coefficient at start (orig. comment; name suggests accel)
VelVis	REAL				False	Set speed (displayed)
O_SegnAcustico	BOOL				False	Horn / acoustic signal
Pot1_Lav	INT				False	Potentiometer 1 value - Work speed -
O_ByPassF	BOOL				False	Front guard bypass

	Pot2_Fin	INT				False	Potentiometer 2 value - Final speed -
	VelVisR	REAL				False	Slow-down speed (displayed)
	ImpEnc	INT				False	Encoder pulses
	SetVelManR	INT				False	% slow-down speed in manual
	MtRal	REAL				False	Slow-down metres
	MtTot	REAL				False	Total metres
	I_StopMetri	BOOL				False	Stop on metre counter
	I_RallEnd	BOOL				False	Slow down at end of roll
	I_FtRall	BOOL				False	Slow-down photocell
	I_RotturaOn	BOOL				False	Paper break monitoring ON
	I_FtRotto	BOOL				False	Paper break photocell
	I_ThAspira	BOOL				False	Extractor fan thermal OK
	RicRsetCnt	BOOL				False	Metre counter reset request
	Psw	INT				False	Password
	I_RsetAlarm	BOOL				False	Alarm reset
	Fix10Vdc_1	UINT				False	Fixed 10 Vdc generator
	DiaRuota	REAL				False	Encoder wheel diameter
	AlarmCodeSV	ARRAY[0..32] OF BOOL				False	Panel alarms
	ProtEscH	BOOL				False	Guards bypassed
	RipaCloseH	BOOL				False	Front guard closed
	MemNocciolo	BOOL				False	(?) Stored core value
	Misurato	REAL				False	Measured distance
	MisuraNocciolo	REAL				False	(?) Core measurement
	AI_Misura	INT				False	Analog value - parent roll measurement
	DiamAlbero	REAL				False	(?) Shaft diameter
	MisuraOn	BOOL				False	(?) Measurement enabled
	DiamBobMis	REAL				False	(?) Measured roll diameter
	DiamVasvaOff	REAL				False	(?) Diameter for tub/cradle off
	DiamStop	REAL				False	(?) Stop diameter
	DiamRall	REAL				False	(?) Slow-down diameter
	H_RsetAlarm	BOOL				False	(?) Alarm reset from panel
	PagPann	ARRAY[0..1] OF DINT				False	HMI panel page
	Psw22	BOOL				False	(?) Password 22
	Psw44	BOOL				False	(?) Password 44
	I_RotoloOk	BOOL				False	(?) Roll OK
	I_FtVascaOff	BOOL				False	(?) Tub/cradle-off photocell
	I_FtEndBob	BOOL				False	(?) End-of-roll photocell

1-7-1-1-2.Generale

0 Remoto 10.146.233.11
Creo un clok di 1s.



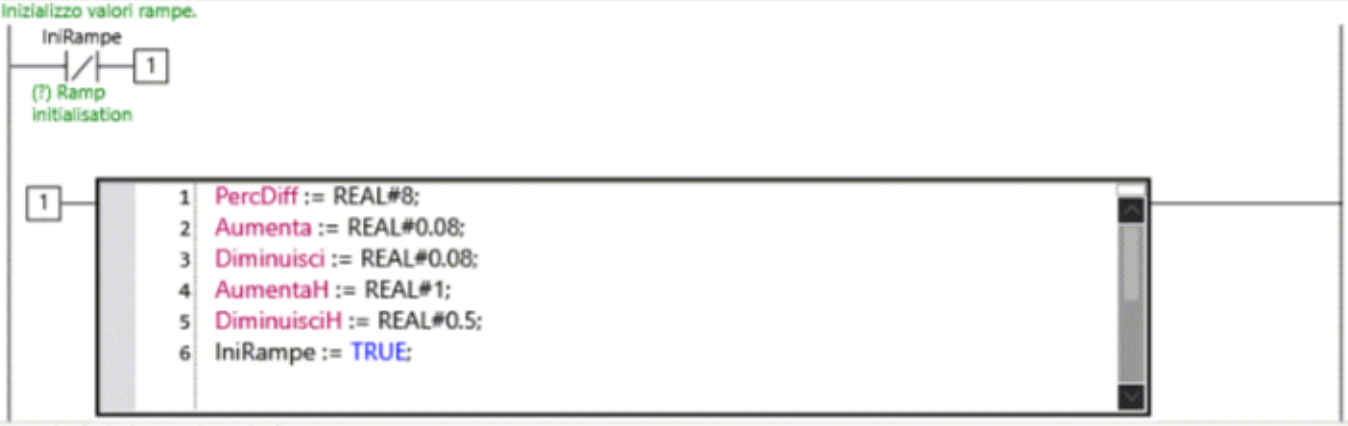
1 Creo un clok di 100ms.



2 Creo un clock di 10ms.



3 Inizializzo valori rampe.

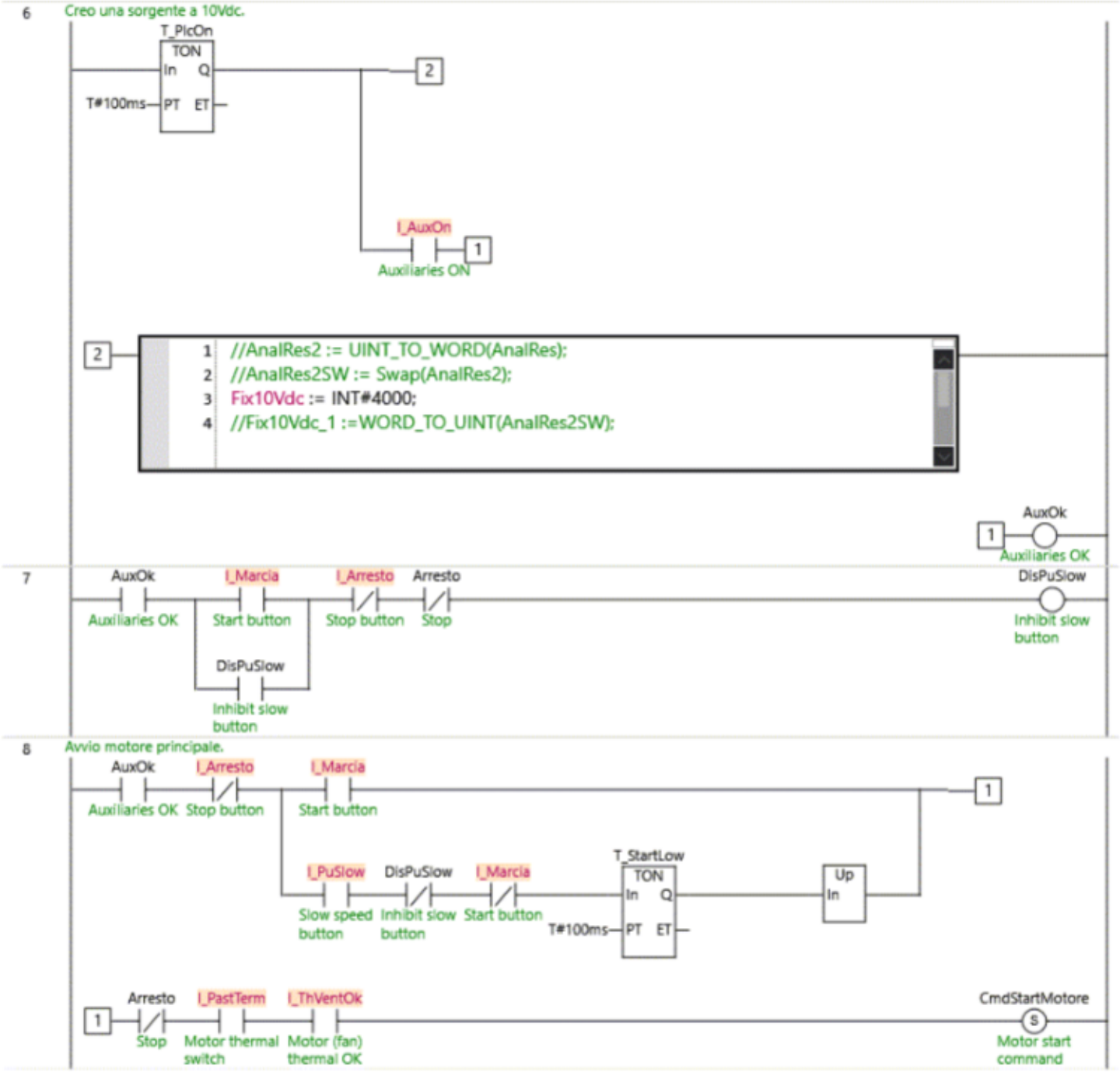


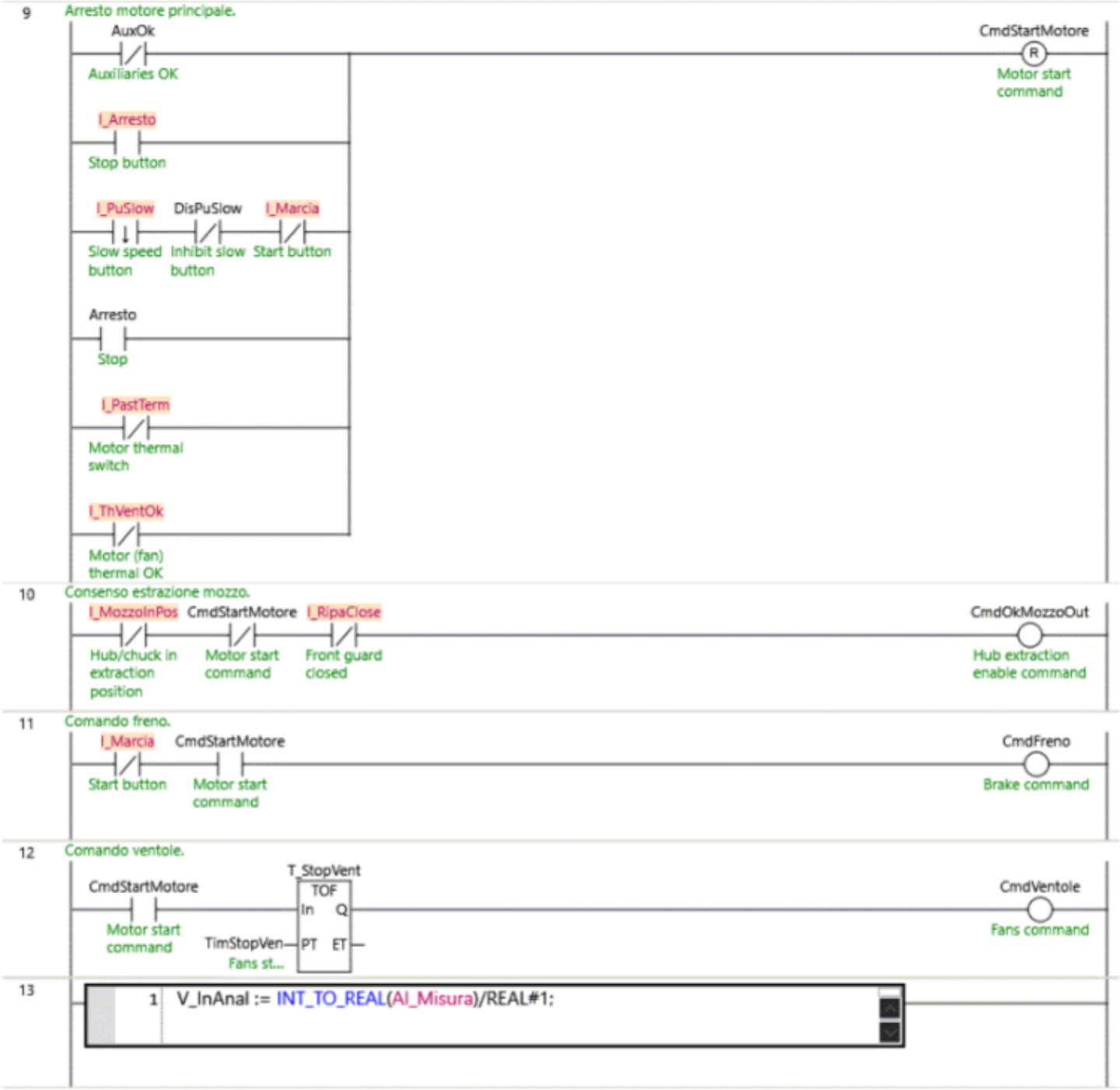
4 Imposto risoluzione scala analogica.

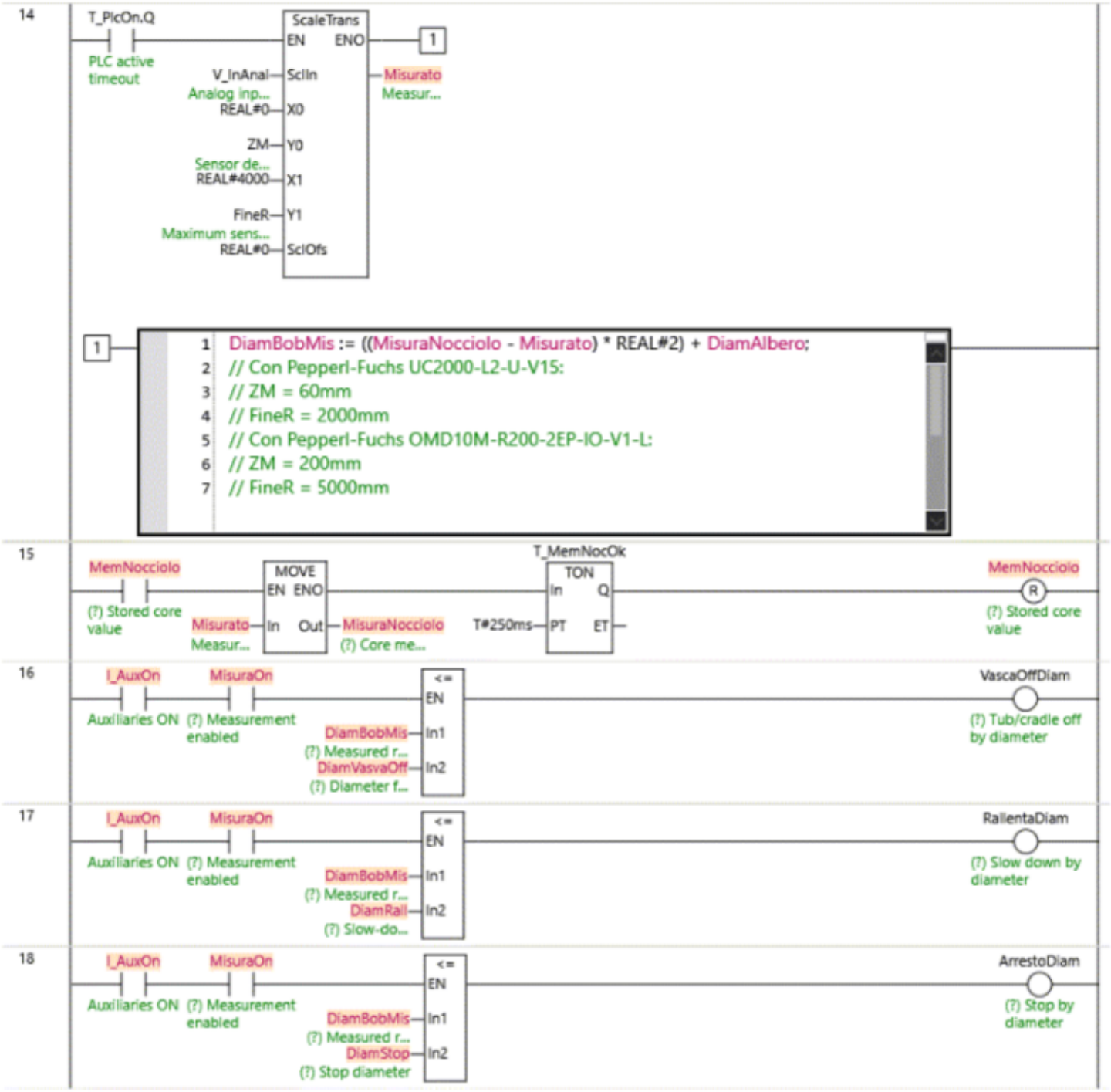


5 Creo variabile per conversioni mt/min --> mm/sec e viceversa.




6
Creo una sorgente a 10Vdc.
T_PlcOn
TON
In Q
T#100ms PT ET
2
I_AuxOn
1
Auxiliaries ON
2
1 //AnalRes2 := UINT_TO_WORD(AnalRes);
2 //AnalRes2SW := Swap(AnalRes2);
3 Fix10Vdc := INT#4000;
4 //Fix10Vdc_1 :=WORD_TO_UINT(AnalRes2SW);
1
AuxOk
Auxiliaries OK
7
AuxOk
Auxiliaries OK
I_Marcia
Start button
DisPuSlow
Inhibit slow button
I_Arresto
Stop button
Arresto
Stop
DisPuSlow
Inhibit slow button
8
Avvio motore principale.
AuxOk
Auxiliaries OK
I_Arresto
Stop button
I_Marcia
Start button
1
I_PuSlow
Slow speed button
DisPuSlow
Inhibit slow button
I_Marcia
Start button
T_StartLow
TON
In Q
T#100ms PT ET
Up
In
1
Arresto
Stop
I_PastTerm
Motor thermal switch
I_ThVentOk
Motor (fan) thermal OK
CmdStartMotore
S
Motor start command


9
Arresto motore principale.
AuxOk
Auxiliaries OK
I_Arresto
Stop button
I_PuSlow
Slow speed button
DisPuSlow
Inhibit slow button
I_Marcia
Start button
Arresto
Stop
I_PastTerm
Motor thermal switch
I_ThVentOk
Motor (fan) thermal OK
CmdStartMotore
R
Motor start command
10
Consenso estrazione mozzo.
I_MozzoInPos
Hub/chuck in extraction position
CmdStartMotore
Motor start command
I_RipaClose
Front guard closed
CmdOkMozzoOut
Hub extraction enable command
11
Comando freno.
I_Marcia
Start button
CmdStartMotore
Motor start command
CmdFreno
Brake command
12
Comando ventole.
CmdStartMotore
Motor start command
T_StopVent
TOF
In
Q
TimStopVen
Fans st...
PT
ET
CmdVentole
Fans command
13
1 V_InAnal := INT_TO_REAL(AI_Misura)/REAL#1;


14
T_PlcOn.Q
PLC active timeout
ScaleTrans
EN
ENO
1
V_InAnal
Analog inp...
REAL#0
SclIn
X0
ZM
Y0
Sensor de...
REAL#4000
X1
FineR
Y1
Maximum sens...
REAL#0
SclOfs
Misurato
Measur...
1
1 DiamBobMis := ((MisuraNocciolo - Misurato) * REAL#2) + DiamAlbero;
2 // Con Pepperl-Fuchs UC2000-L2-U-V15:
3 // ZM = 60mm
4 // FineR = 2000mm
5 // Con Pepperl-Fuchs OMD10M-R200-2EP-IO-V1-L:
6 // ZM = 200mm
7 // FineR = 5000mm
15
MemNocciolo
(?) Stored core value
MOVE
EN ENO
Misurato
Measur...
In
Out
MisuraNocciolo
(?) Core me...
T_MemNocOk
TON
In
Q
T#250ms
PT
ET
MemNocciolo
R
(?) Stored core value
16
I_AuxOn
Auxiliaries ON
MisuraOn
(?) Measurement enabled
<=
EN
DiamBobMis
(?) Measured r...
In1
DiamVasvaOff
(?) Diameter f...
In2
VascaOffDiam
(?) Tub/cradle off by diameter
17
I_AuxOn
Auxiliaries ON
MisuraOn
(?) Measurement enabled
<=
EN
DiamBobMis
(?) Measured r...
In1
DiamRall
(?) Slow-do...
In2
RallentaDiam
(?) Slow down by diameter
18
I_AuxOn
Auxiliaries ON
MisuraOn
(?) Measurement enabled
<=
EN
DiamBobMis
(?) Measured r...
In1
DiamStop
(?) Stop diameter
In2
ArrestoDiam
(?) Stop by diameter

1-7-1-1-3.GestioneMotore

0 Abilito la lettura dell'encoder.



1 Calcolo mm per impulso encoder.

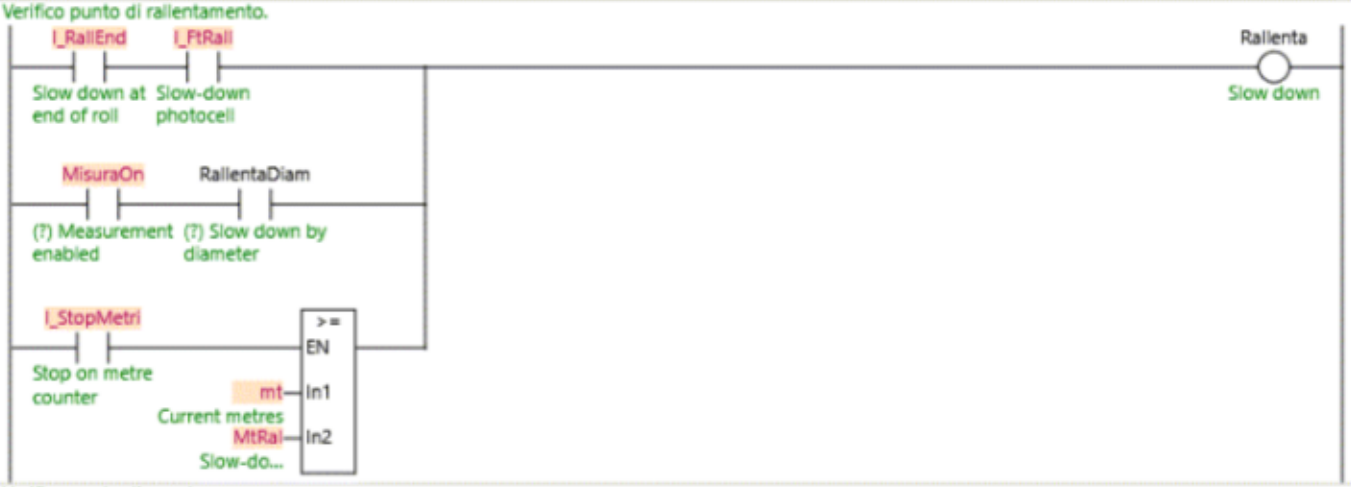
```
IF ImpEnc = INT#0 THEN;
    ImpEnc :=INT#1;
END_IF;
//Diametro standard ruota 400mm
IF  DiaRuota = REAL#0 THEN;
    DiaRuota := REAL#400;
END_IF;

mm_Imp := DiaRuota / (INT_TO_REAL(ImpEnc) * REAL#4);
```

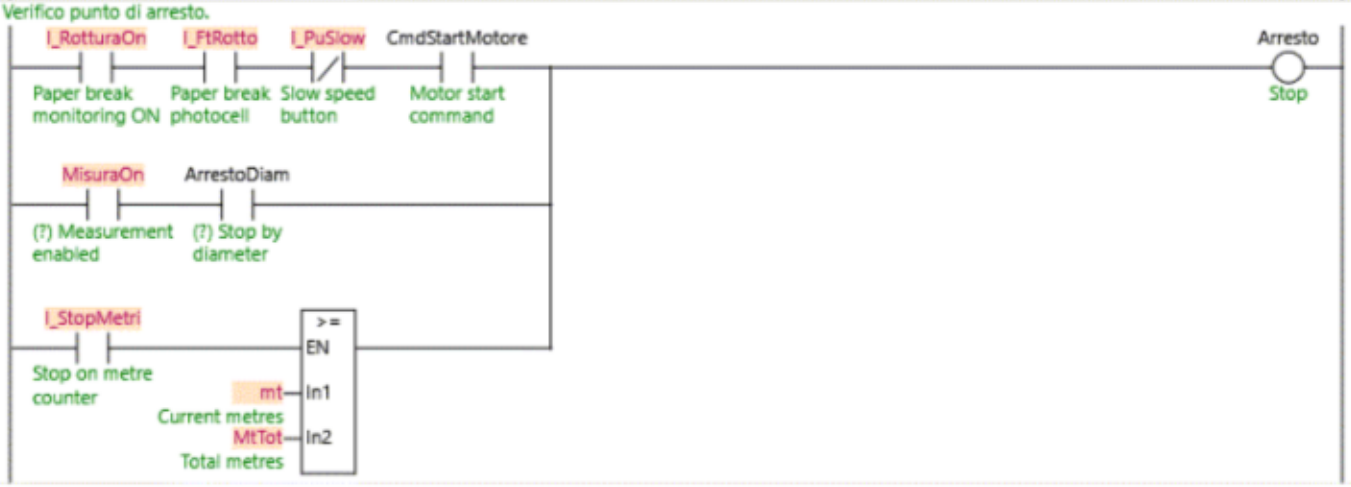
2 Reset contatore encoder.



3 Verifico punto di rallentamento.



4 Verifico punto di arresto.



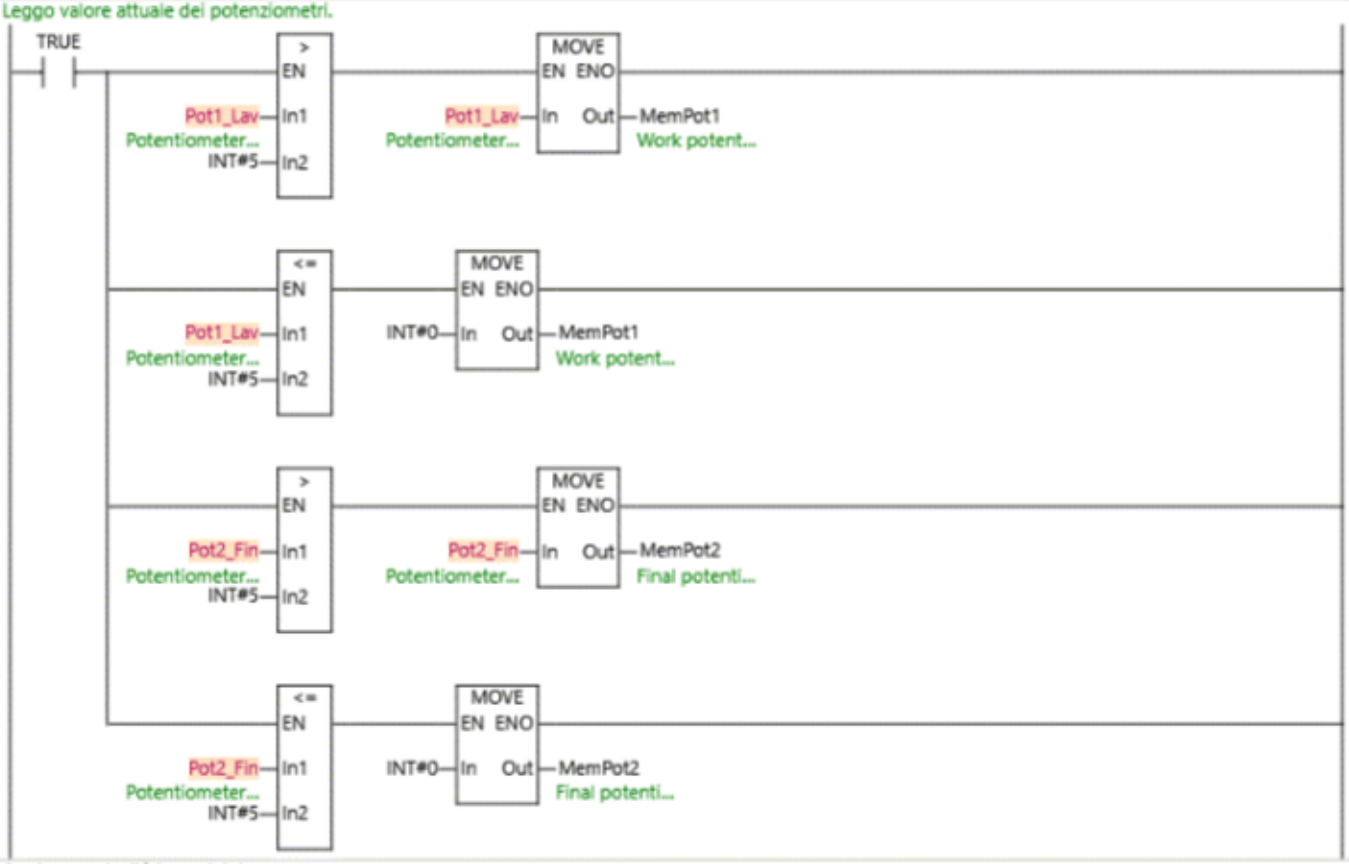
5 Verifico il valore massimo e minimo di velocità lavoro e lenta.

```
IF VelMax < REAL#600 THEN;
    VelMax := REAL#600;
ELSIF VelMax > REAL#1000 THEN;
    VelMax := REAL#1000;
END_IF;
VelMax_mms := VelMax * REAL#1000 / REAL#60;

IF VelSetMan < REAL#10 THEN;
    VelSetMan := REAL#10;
ELSIF VelSetMan > REAL#70 THEN;
    VelSetMan := REAL#70;
END_IF;

RappAnalOut := (VelMax * Mt_To_mm) / UINT_TO_REAL(AnalRes); //REAL#4000;
RappAnalIn := UINT_TO_REAL(AnalRes) / VelMax ;
```

6 Leggo valore attuale dei potenziometri.

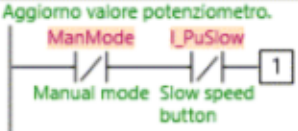


7 Aggiorno velocità impostata.

```
IF ManMode = FALSE THEN;
    VelVis := UINT_TO_REAL(MemPot1) / RappAnalIn;
    VelVisR := UINT_TO_REAL(MemPot2) / RappAnalIn;
ELSE;
    VelVis := MemPot1 / (AnalRes  /UINT#40); // INT#40;
    VelVisR := MemPot2 / (AnalRes / UINT#40); // INT#40;

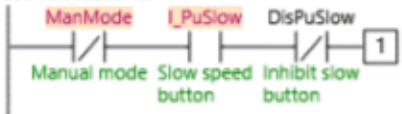
END_IF;
```

8 Aggiorno valore potenziometro.




```
IF Rallenta = FALSE THEN;
    VelSet := UINT_TO_REAL(MemPot1) / RappAnalIn;
ELSE;
    VelSet := UINT_TO_REAL(MemPot2) / RappAnalIn;
END_IF;
VelSet_mms := VelSet * Mt_To_mm;
```

9 Uso velocità lenta.




```
VelSet := VelSetMan;
VelSet_mms := VelSet * Mt_To_mm;
```

10 Verifico differenza tra velocità impostata e quella rilevata.

```
//Diff := VelSet_mms * PercDiff / REAL#100;
Diff := PercDiff;
DiffMax := VelSet + Diff;
DiffMin := VelSet - Diff;
// Troppo veloce.
IF Vel_mtmin > DiffMax THEN;
    H_Range := TRUE;
ELSE
    H_Range := FALSE;
END_IF;
// Troppo lento.
IF Vel_mtmin < DiffMin THEN;
    L_Range := TRUE;
ELSE
    L_Range := FALSE;
END_IF;
```

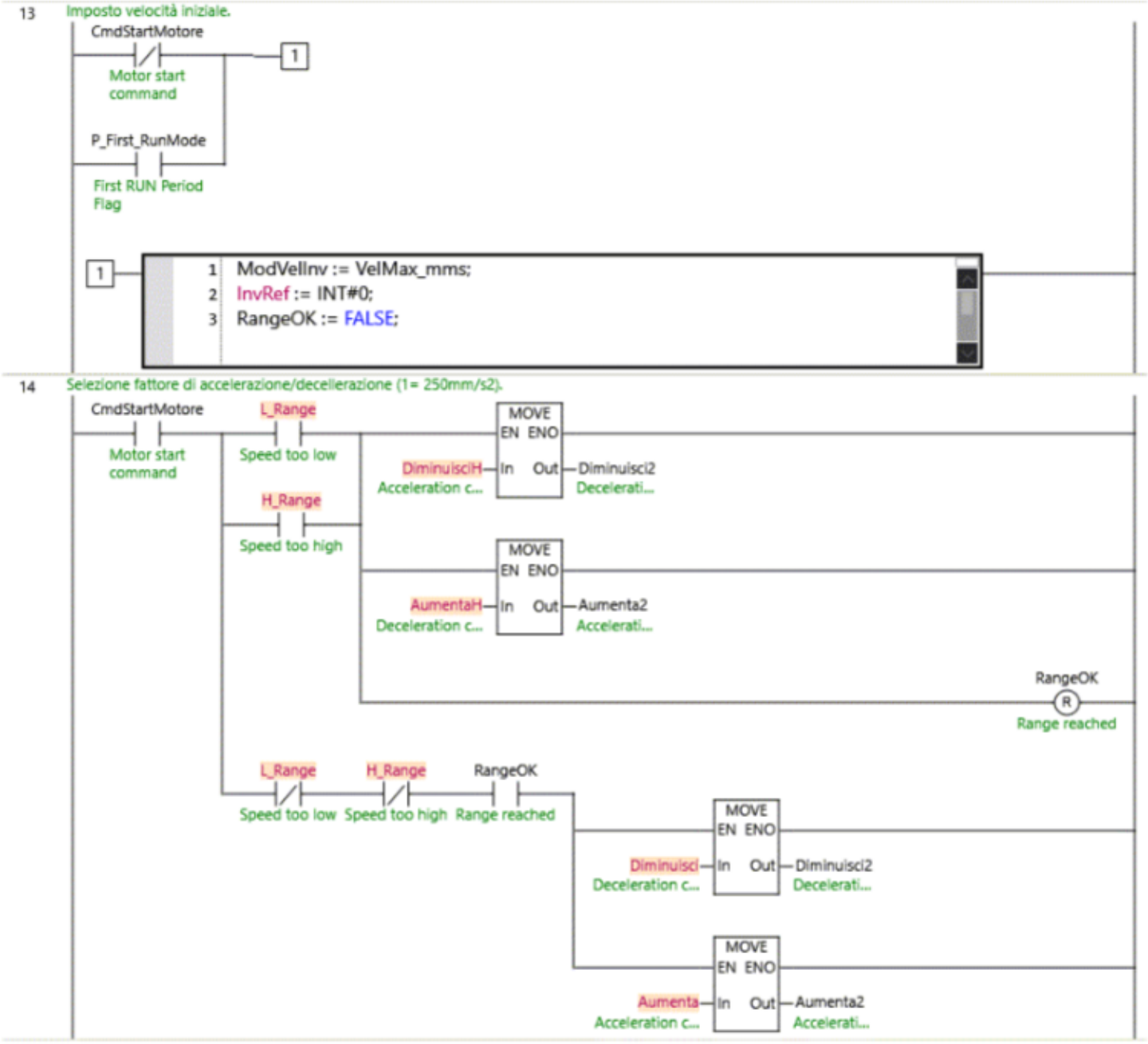
11 Leggo il valore attuale dell'encoder.



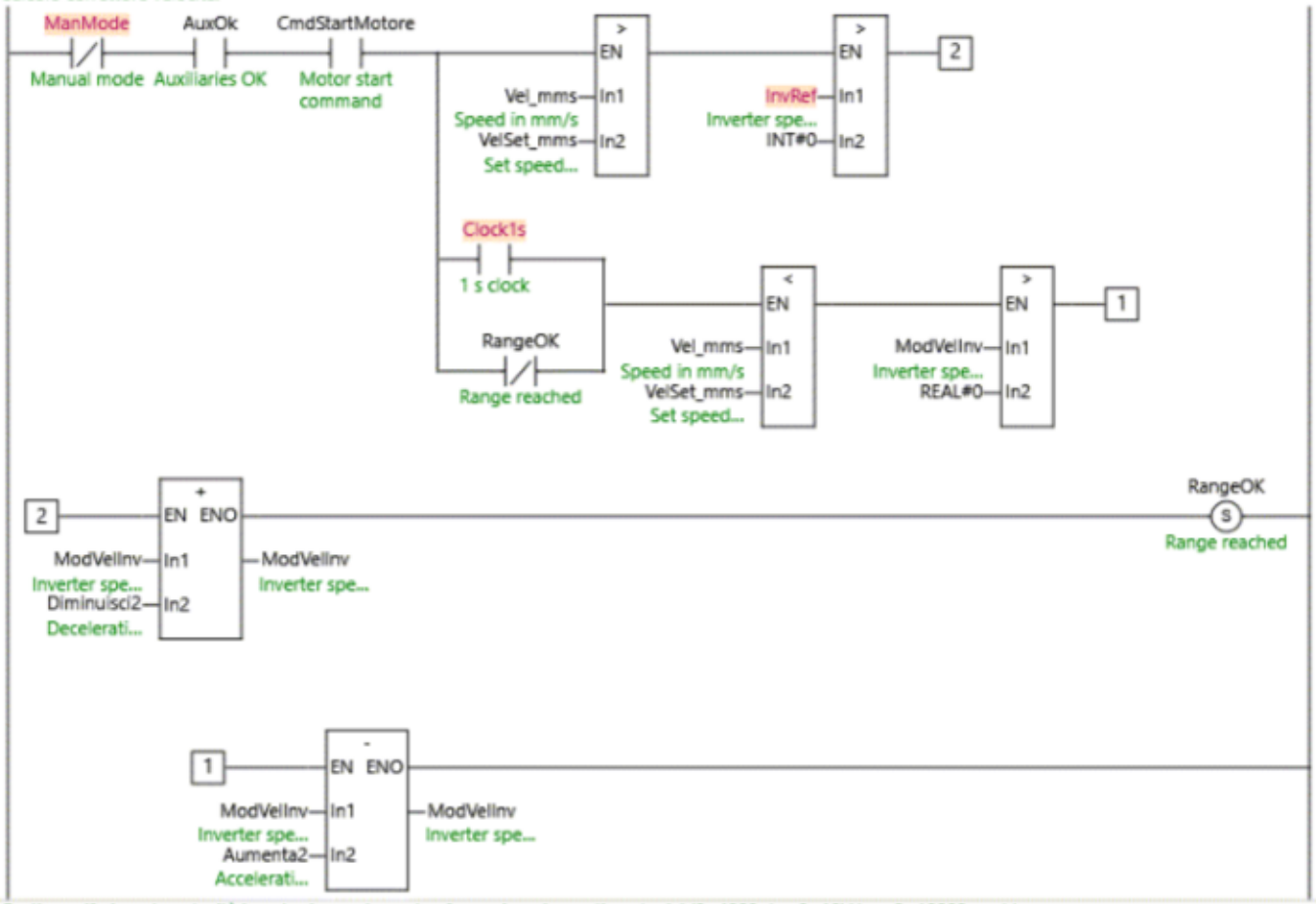
12 Calcolo velocità.



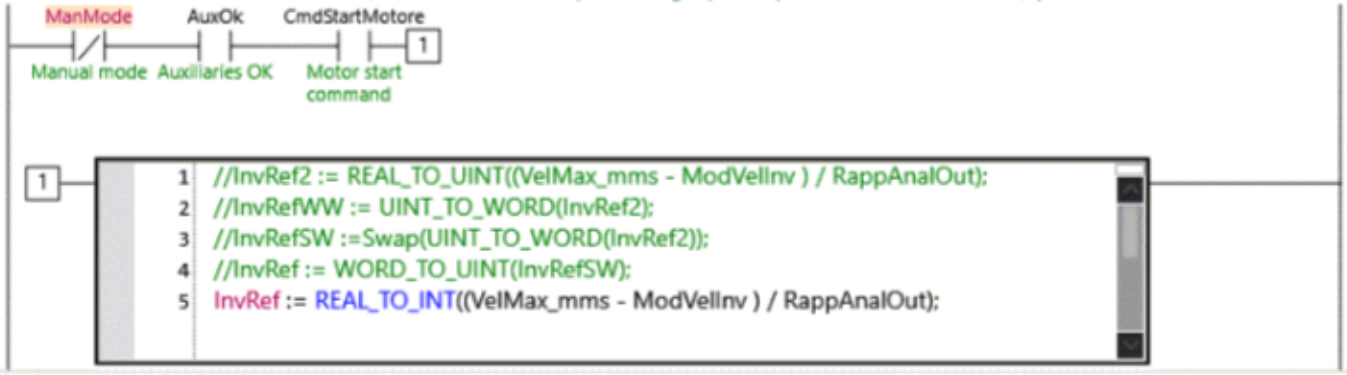

```
DeltaPulse := EncPos - MemPos;
Vel_mms := (DINT_TO_REAL(DeltaPulse) * mm_Imp) * REAL#20; // Campionamento ogni 1/20sec. (50ms)
Vel_mtmin := Vel_mms / Mt_To_mm;
```


13
Imposto velocità iniziale.
CmdStartMotore
Motor start command
1
P_First_RunMode
First RUN Period Flag
1
1 ModVelInv := VelMax_mms;
2 InvRef := INT#0;
3 RangeOK := FALSE;
14
Selezione fattore di accelerazione/decelerazione (1= 250mm/s2).
CmdStartMotore
Motor start command
L_Range
Speed too low
MOVE
EN ENO
DiminuisciH
Acceleration c...
In Out
Diminuisci2
Decelerati...
H_Range
Speed too high
MOVE
EN ENO
AumentaH
Deceleration c...
In Out
Aumenta2
Accelerati...
RangeOK
R
Range reached
L_Range
Speed too low
H_Range
Speed too high
RangeOK
Range reached
MOVE
EN ENO
Diminuisci
Deceleration c...
In Out
Diminuisci2
Decelerati...
MOVE
EN ENO
Aumenta
Acceleration c...
In Out
Aumenta2
Accelerati...

15 Calcolo correttore velocità.



16 Gestione riferimento velocità inverter in mm/sec e trasformazione in punti analogici (0÷4000pt = 0÷10Vdc = 0÷10000mm/s).

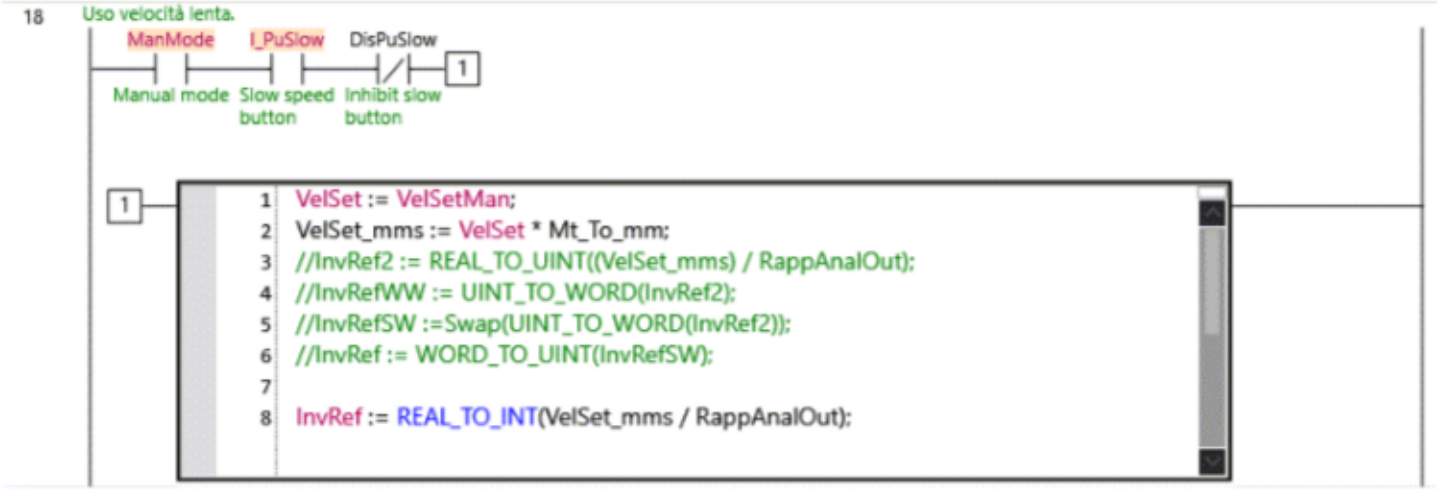



```
1  //InvRef2 := REAL_TO_UINT((VelMax_mms - ModVelInv ) / RappAnalOut);
2  //InvRefWW := UINT_TO_WORD(InvRef2);
3  //InvRefSW :=Swap(UINT_TO_WORD(InvRef2));
4  //InvRef := WORD_TO_UINT(InvRefSW);
5  InvRef := REAL_TO_INT((VelMax_mms - ModVelInv ) / RappAnalOut);
```

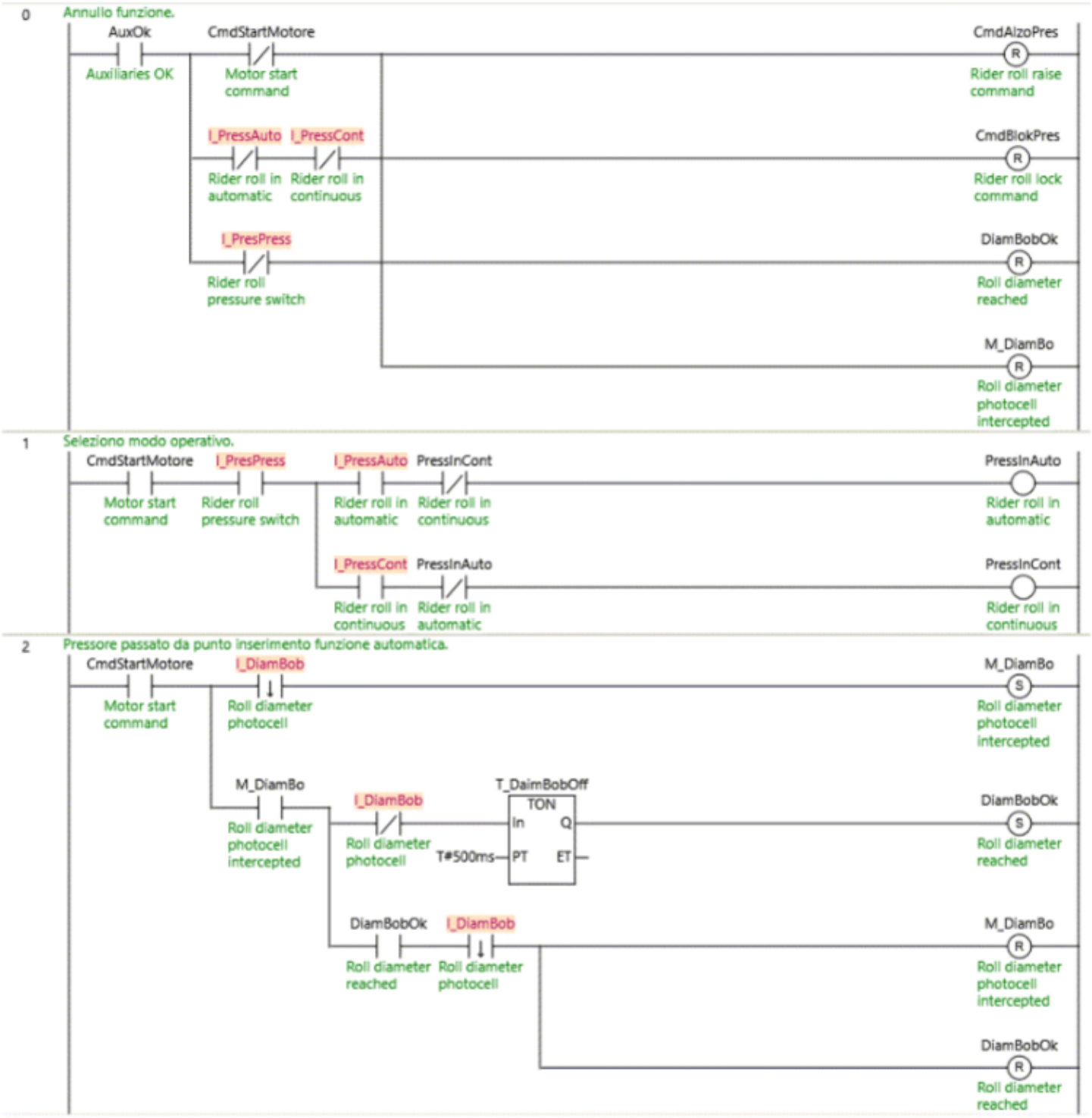
17 Aggiorno valore potenziometro.

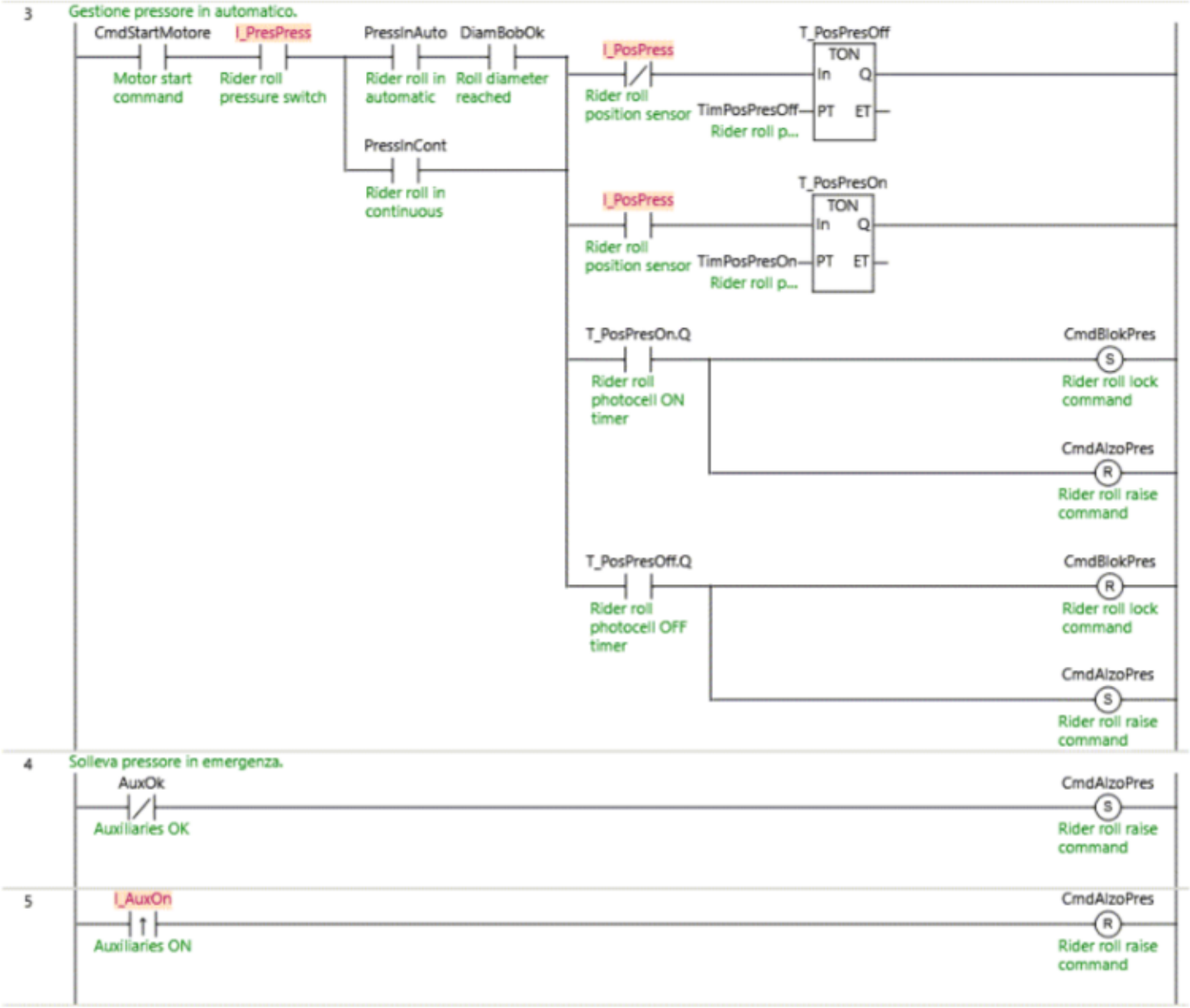



```
1  IF Rallenta = FALSE THEN;
2      InvRef := Pot1_Lav;
3  ELSE;
4      InvRef := Pot2_Fin;
5  END_IF;
```


18
Uso velocità lenta.
ManMode
I_PuSlow
DisPuSlow
1
Manual mode
Slow speed button
Inhibit slow button
1
1 VelSet := VelSetMan;
2 VelSet_mms := VelSet * Mt_To_mm;
3 //InvRef2 := REAL_TO_UINT((VelSet_mms) / RappAnalOut);
4 //InvRefWW := UINT_TO_WORD(InvRef2);
5 //InvRefSW :=Swap(UINT_TO_WORD(InvRef2));
6 //InvRef := WORD_TO_UINT(InvRefSW);
7
8 InvRef := REAL_TO_INT(VelSet_mms / RappAnalOut);

1-7-1-1-4.Pressore




3
Gestione pressore in automatico.
CmdStartMotore
Motor start command
I_PresPress
Rider roll pressure switch
PressInAuto
Rider roll in automatic
DiamBobOk
Roll diameter reached
PressInCont
Rider roll in continuous
I_PosPress
Rider roll position sensor
T_PosPresOff
TON
In
Q
PT
ET
TimPosPresOff
Rider roll p...
I_PosPress
Rider roll position sensor
T_PosPresOn
TON
In
Q
PT
ET
TimPosPresOn
Rider roll p...
T_PosPresOn.Q
Rider roll photocell ON timer
CmdBlokPres
S
Rider roll lock command
CmdAlzoPres
R
Rider roll raise command
T_PosPresOff.Q
Rider roll photocell OFF timer
CmdBlokPres
R
Rider roll lock command
CmdAlzoPres
S
Rider roll raise command
4
Solleva pressore in emergenza.
AuxOk
Auxiliaries OK
CmdAlzoPres
S
Rider roll raise command
5
I_AuxOn
Auxiliaries ON
CmdAlzoPres
R
Rider roll raise command

1-7-1-1-5.Pannello

0 Creo variabili di tipo 'TIME'.

```
1  TimStopVen := NanoSecToTime(REAL_TO_LINT(TimStopVenP * REAL#60000000000));
2  TimPosPresOn := NanoSecToTime(REAL_TO_LINT(TimPosPresOnP *
   REAL#1000000000));
3  TimPosPresOff := NanoSecToTime(REAL_TO_LINT(TimPosPresOffP *
   REAL#1000000000));
```

1 Converto gli impulsi in mm e mt.

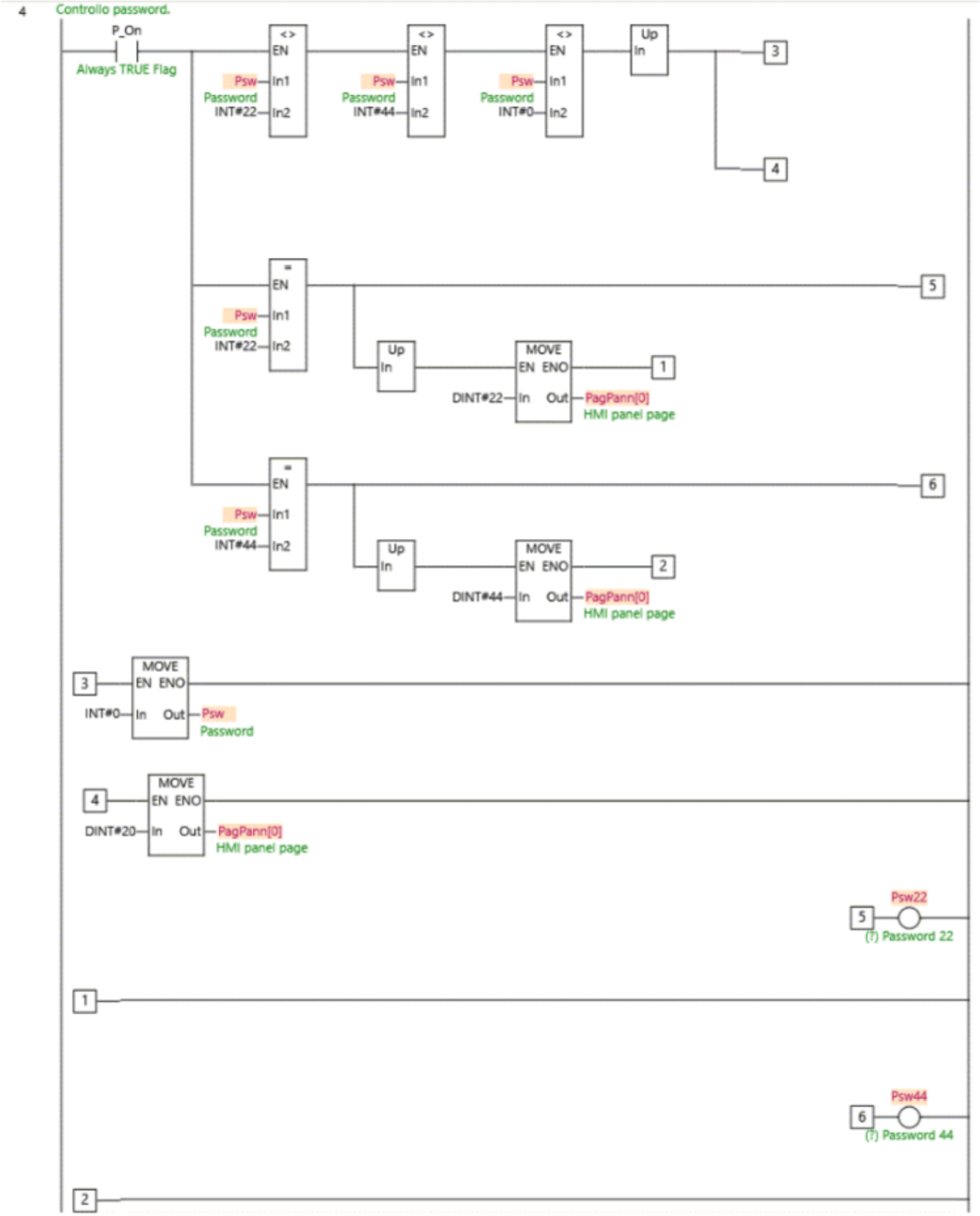
```
1  mm := DINT_TO_REAL(EncPos) * mm_Imp;
2  mt := mm / REAL#1000;
```

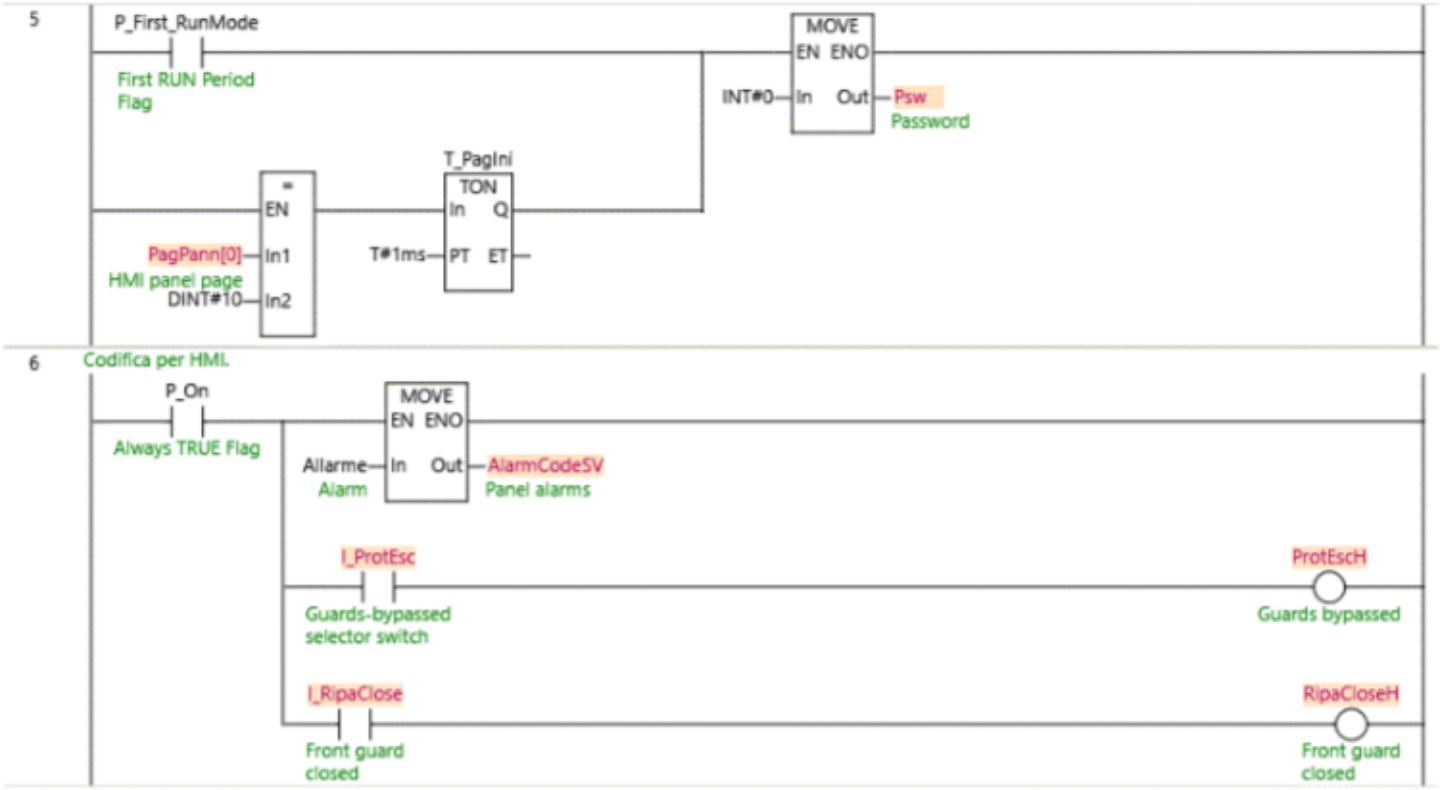
2



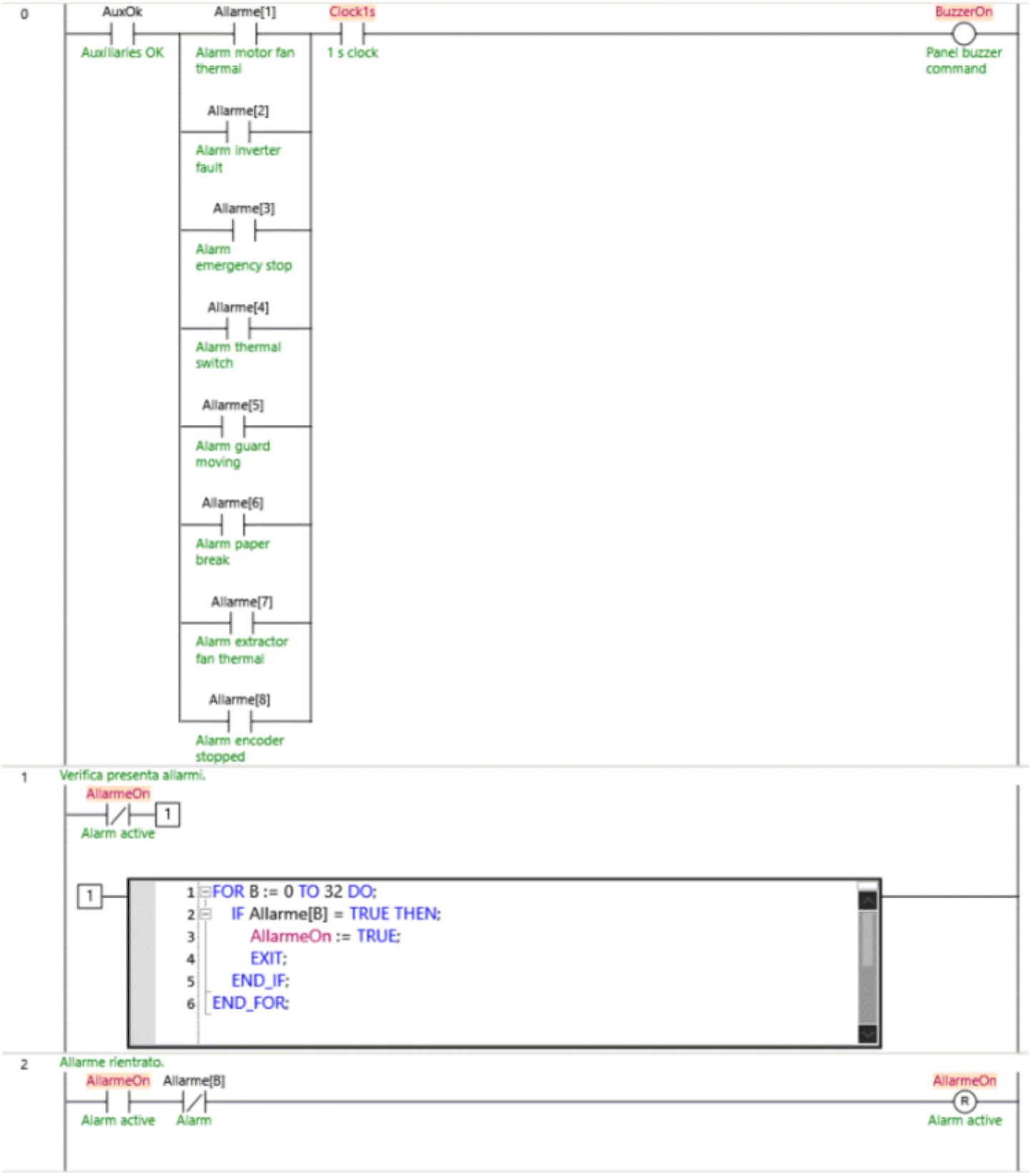
3 Richiesta azzeramento contametri.




4
Controllo password.
P_On
Always TRUE Flag
<>
EN
In1
In2
Psw
Password
INT#22
INT#44
INT#0
Up
In
3
4
=
5
6
MOVE
EN ENO
In Out
DINT#22
DINT#44
PagPann[0]
HMI panel page
1
2
INT#0
Psw
Password
DINT#20
Psw22
(?) Password 22
Psw44
(?) Password 44


5
P_First_RunMode
First RUN Period Flag
MOVE
EN ENO
INT#0
In
Out
Psw
Password
T_PagIni
TON
In Q
T#1ms
PT ET
=
EN
PagPann[0]
In1
HMI panel page
DINT#10
In2
6
Codifica per HMI
P_On
Always TRUE Flag
MOVE
EN ENO
Allarme
Alarm
In
Out
AlarmCodeSV
Panel alarms
I_ProtEsc
Guards-bypassed selector switch
ProtEscH
Guards bypassed
I_RipaClose
Front guard closed
RipaCloseH
Front guard closed

1-7-1-1-6.Allarmi



3 Termico ventola motore.



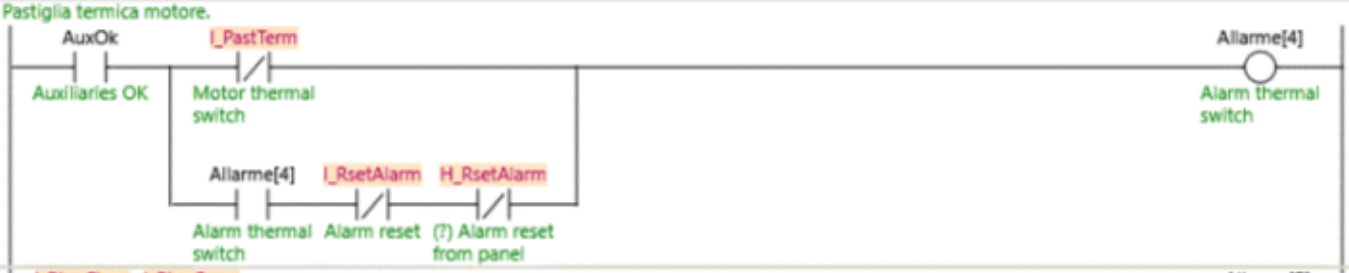
4 Anomalia inverter.



5 Macchina in emergenza.



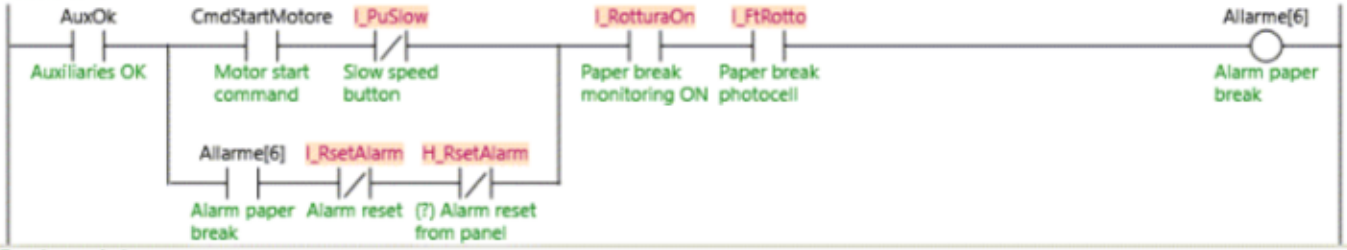
6 Pastiglia termica motore.



7



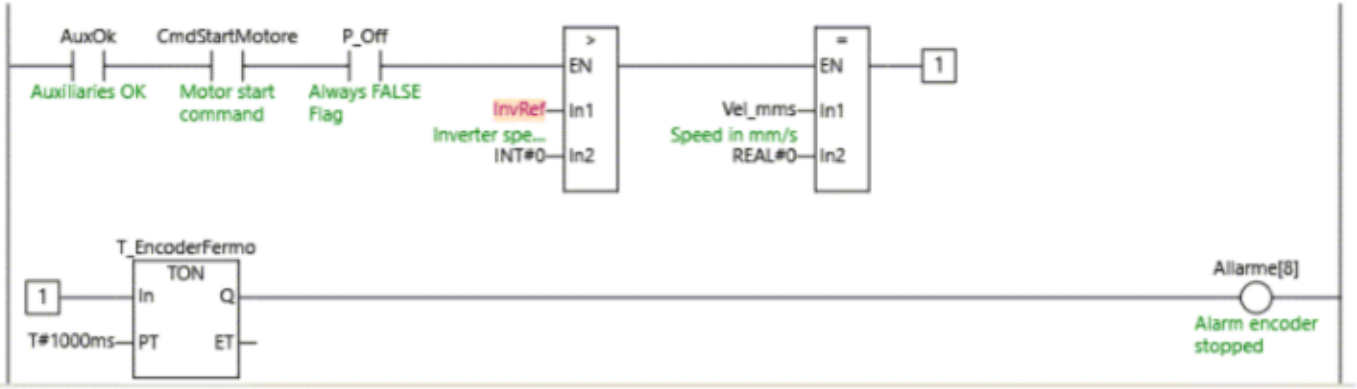
8 Rottura carta.



9 Termico aspiratore.



10 Encoder fermo.



11 Giunto portarotolo non chiuso.



12 Mozzo non in posizione.



13 Fotocellula vasca off.



14 Micro a fune attivo.



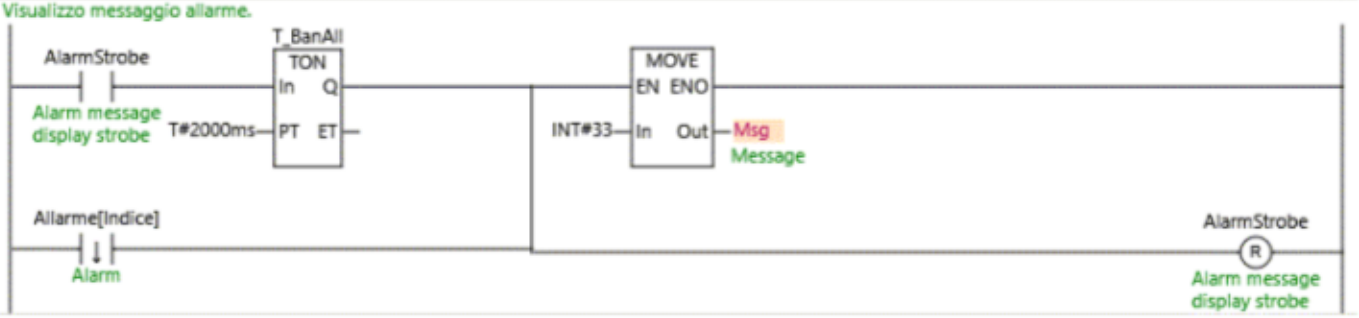
15 Fotocellula rallentamento attiva.



16 Fotocellula fine bobina attiva.



17 Visualizzo messaggio allarme.



18 Strobe visualizza messagio allarme.



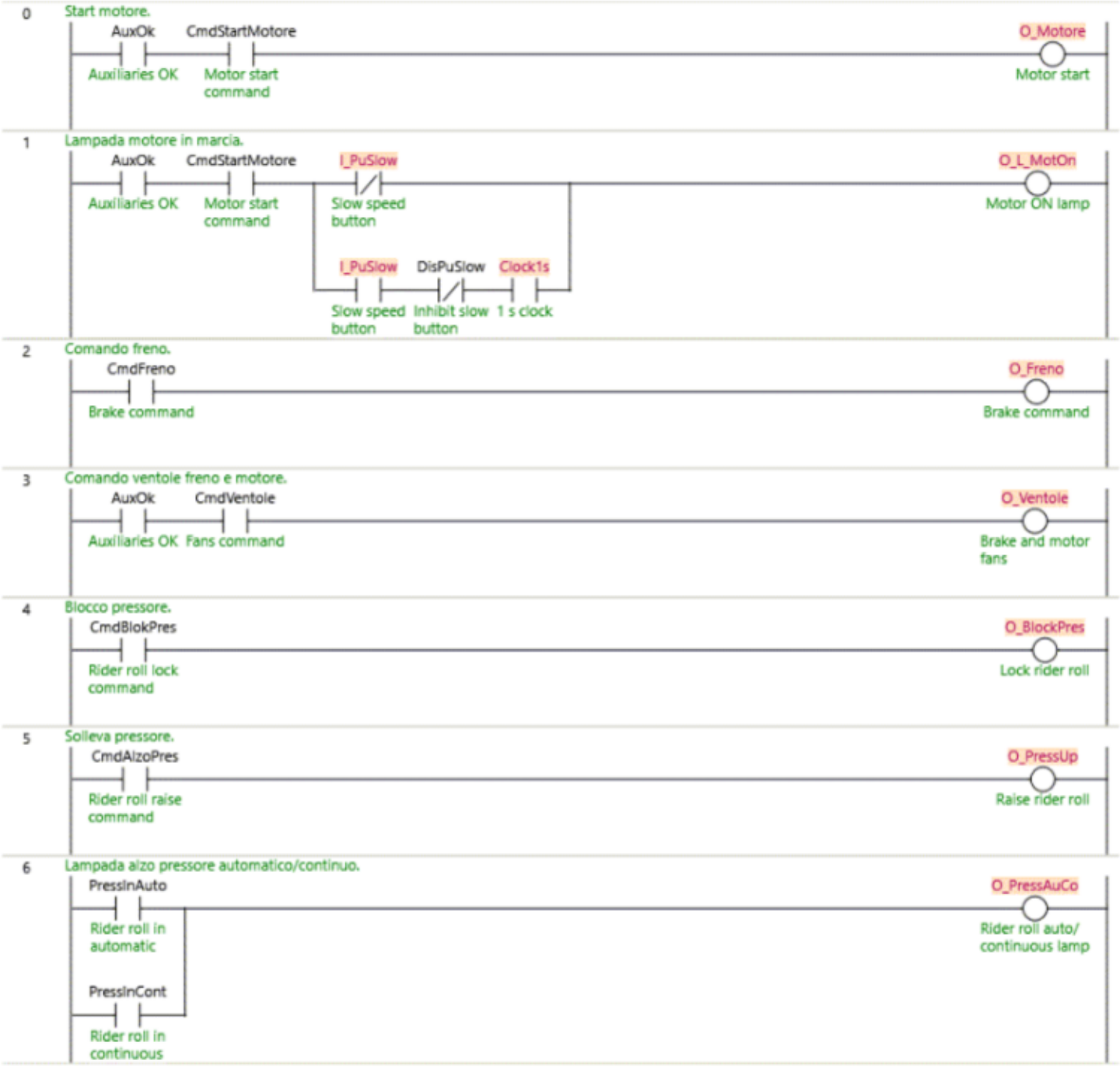
19 Ciclo controllio presenza allarme.



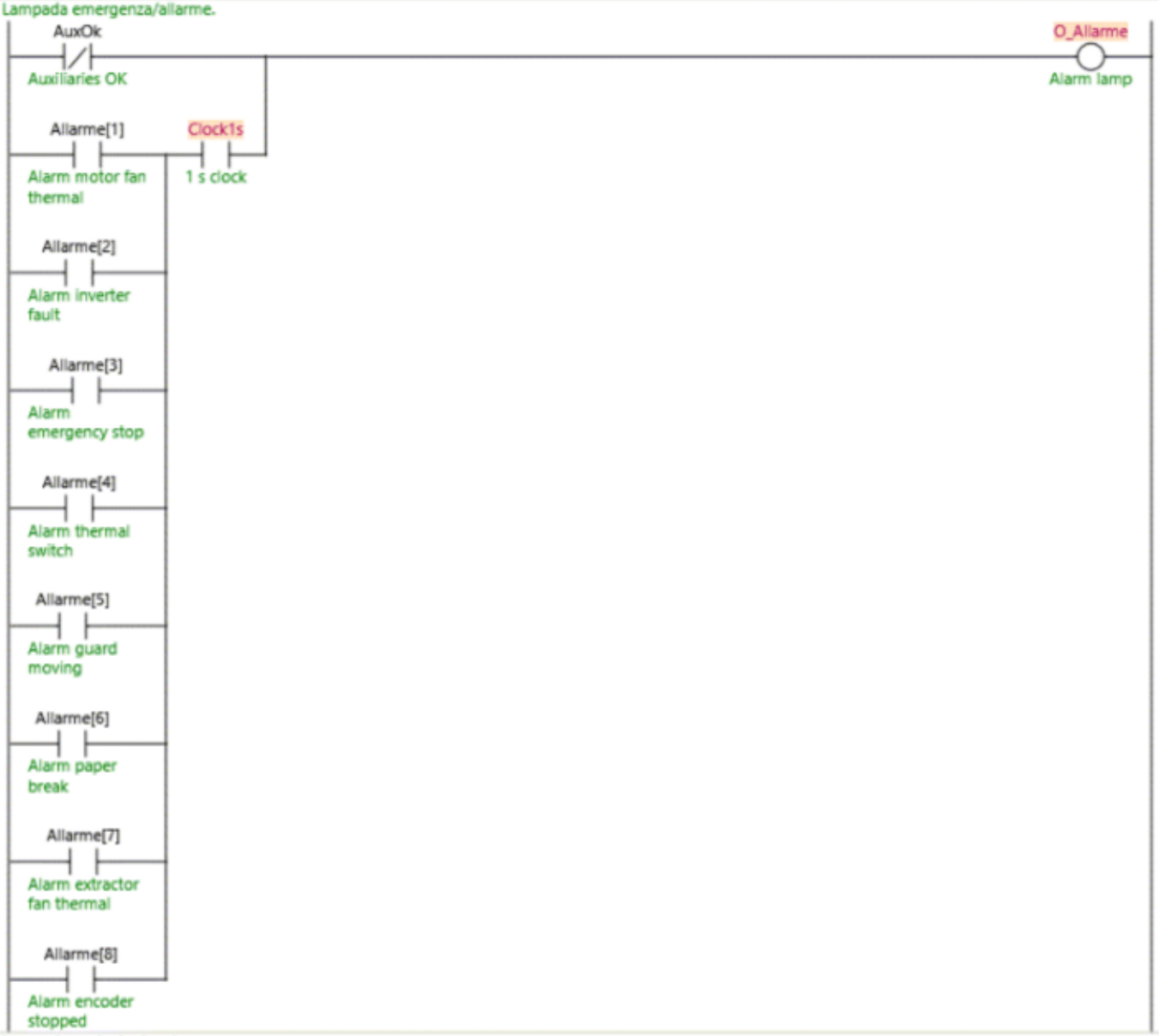
20 Inizializzo messaggio allarme.



1-7-1-1-7.GestioneUscite



7 Lampada emergenza/allarme.



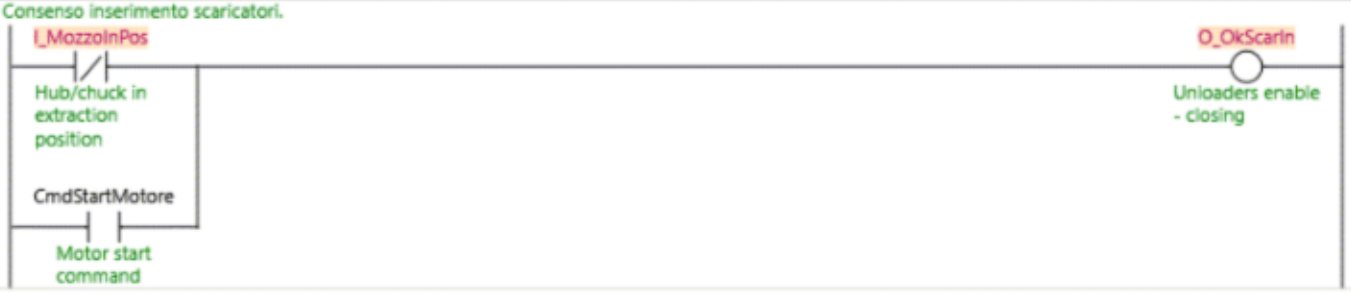
8 Lampada protezioni escluse.

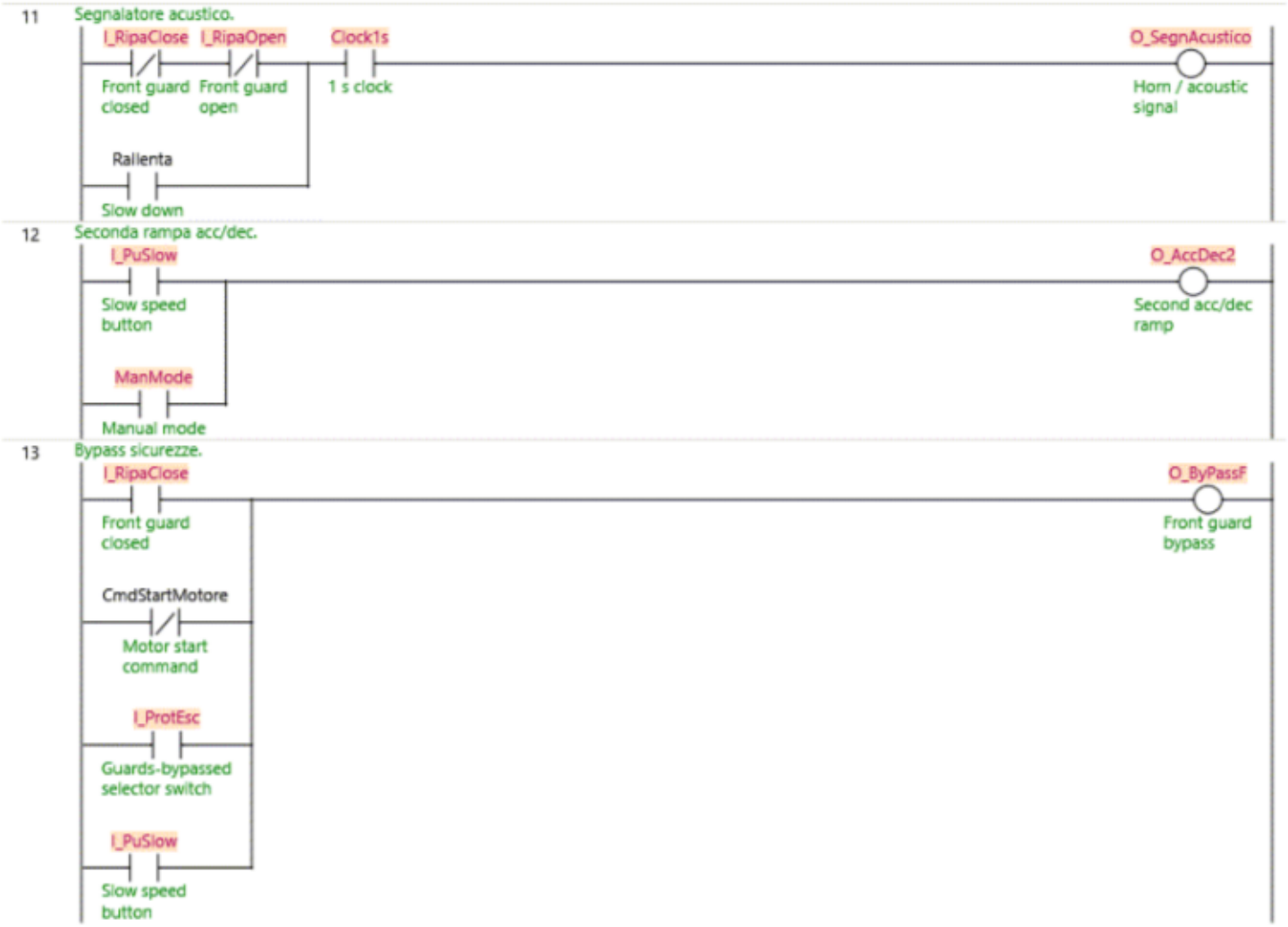


9 Consenso estrazione mozzo.



10 Consenso inserimento scaricatori.




11
Segnalatore acustico.
I_RipaClose
Front guard closed
I_RipaOpen
Front guard open
Clock1s
1 s clock
O_SegnAcustico
Horn / acoustic signal
Rallenta
Slow down
12
Seconda rampa acc/dec.
I_PuSlow
Slow speed button
O_AccDec2
Second acc/dec ramp
ManMode
Manual mode
13
Bypass sicurezze.
I_RipaClose
Front guard closed
O_ByPassF
Front guard bypass
CmdStartMotore
Motor start command
I_ProtEsc
Guards-bypassed selector switch
I_PuSlow
Slow speed button

1-8.Data

1-8-1.Data Types

Name	Type	Comment

1-8-2.Global Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Network Publish	Comment
VAR_GLOBAL							
Pot1_Lav	INT		BuiltInIO://opt/#0/Ch1_Analog_Input_Value	False	False	Publish Only	Potentiometer 1 value - Work speed -
InvRef	INT		BuiltInIO://opt/#0/Ch2_Analog_Output_Value	False	False	Publish Only	Inverter speed reference
Fix10Vdc	INT		BuiltInIO://opt/#0/Ch1_Analog_Output_Value	False	False	Publish Only	Fixed 10 Vdc generator
VelMax	REAL			True	False	Publish Only	Maximum speed
VelSet	REAL			False	False	Publish Only	Set speed
H_Range	BOOL			False	False	Publish Only	Speed too high
L_Range	BOOL			False	False	Publish Only	Speed too low
PercDiff	REAL			True	False	Publish Only	Speed difference percentage
EncPos	DINT		IOBus://unit#3/Ch1 Encoder Present Position	False	False	Publish Only	Encoder counter
PulseRate	UDINT		IOBus://unit#3/Ch1 Pulse Rate	False	False	Publish Only	Pulses per 100 ms
EncOn	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Counter Enable	False	False	Publish Only	Encoder enable
EncRset	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Internal Reset Execution	False	False	Publish Only	Encoder reset
PulsPerOn	BOOL		IOBus://unit#3/Ch1 Pulse Period Measurement Function/Ch1 Pulse Period Measurement Enable	False	False	Publish Only	Enable pulse counter per 100 ms
PulsPerRset	BOOL		IOBus://unit#3/Ch1 Pulse Period Measurement Function/Ch1 Pulse Period Measurement Value Clear	False	False	Publish Only	Reset pulse counter per 100 ms
PagPann	ARRAY[0..1] O			False	False	Publish Only	HMI panel page

	F DINT						
Vel_mtmin	REAL			True	False	Publish Only	Speed in m/min
mt	REAL			True	False	Publish Only	Current metres
I_RsetCnt	BOOL			False	False	Publish Only	Metre counter reset button
VelVis	REAL			True	False	Publish Only	Set speed (displayed)
I_AuxOn	BOOL		BuiltInIO://cpu/#0/Input_Bit_00	False	False	Publish Only	Auxiliaries ON
I_Marcia	BOOL		BuiltInIO://cpu/#0/Input_Bit_01	False	False	Publish Only	Start button
I_Arresto	BOOL		BuiltInIO://cpu/#0/Input_Bit_02	False	False	Publish Only	Stop button
I_PastTerm	BOOL		BuiltInIO://cpu/#0/Input_Bit_03	False	False	Publish Only	Motor thermal switch
I_ProtEsc	BOOL		BuiltInIO://cpu/#0/Input_Bit_04	False	False	Publish Only	Guards-bypassed selector switch
I_MozzoInPos	BOOL		BuiltInIO://cpu/#0/Input_Bit_05	False	False	Publish Only	Hub/chuck in extraction position
I_RipaClose	BOOL		BuiltInIO://cpu/#0/Input_Bit_06	False	False	Publish Only	Front guard closed
I_PuSlow	BOOL		BuiltInIO://cpu/#0/Input_Bit_07	False	False	Publish Only	Slow speed button
I_InvAll	BOOL		BuiltInIO://cpu/#0/Input_Bit_08	False	False	Publish Only	Inverter in alarm
I_PorteClose	BOOL			False	False	Publish Only	Side doors closed
I_BarOk	BOOL			False	False	Publish Only	Light barriers reset
I_DiamBob	BOOL		BuiltInIO://cpu/#0/Input_Bit_13	False	False	Publish Only	Roll diameter photocell
I_PosPress	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN01 (Pin 1.1) _6000_01	False	False	Publish Only	Rider roll position sensor
I_PresPress	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN02 (Pin 2.1) _6000_02	False	False	Publish Only	Rider roll pressure switch
I_PressAuto	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN03 (Pin 1.2) _6000_03	False	False	Publish Only	Rider roll in automatic
I_PressCont	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_I	False	False	Publish Only	Rider roll in continuous

			N04 (Pin 2.2) _6000_04				
O_Motore	BOOL		BuiltInIO://cpu/#0/Output_Bit_00	False	False	Publish Only	Motor start
I_ThVentOk	BOOL		ECAT://node #[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN06 (Pin 2.3) _6000_06	False	False	Publish Only	Motor (fan) thermal OK
O_Allarme	BOOL		ECAT://node #[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 01 (Pin 1.1) _7000_01	False	False	Publish Only	Alarm lamp
O_ProtEsclus e	BOOL			False	False	Publish Only	Guards bypassed lamp
O_PorteOpen	BOOL			False	False	Publish Only	Doors open lamp
O_L_MotOn	BOOL		BuiltInIO://cpu/#0/Output_Bit_01	False	False	Publish Only	Motor ON lamp
O_OkScaricat ori	BOOL		ECAT://node #[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 03 (Pin 1.2) _7000_03	False	False	Publish Only	Unloaders enable - opening
O_ByPassF	BOOL		ECAT://node #[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 04 (Pin 2.2) _7000_04	False	False	Publish Only	Front guard bypass
O_Freno	BOOL		BuiltInIO://cpu/#0/Output_Bit_02	False	False	Publish Only	Brake command
O_PressAuC o	BOOL		BuiltInIO://cpu/#0/Output_Bit_06	False	False	Publish Only	Rider roll auto/continuous lamp
O_PressUp	BOOL		BuiltInIO://cpu/#0/Output_Bit_09	False	False	Publish Only	Raise rider roll
O_BlockPres	BOOL		BuiltInIO://cpu/#0/Output_Bit_04	False	False	Publish Only	Lock rider roll
O_Ventole	BOOL		BuiltInIO://cpu/#0/Output_Bit_03	False	False	Publish Only	Brake and motor fans
TimStopVenP	REAL			True	False	Publish Only	Fans stop timeout (from panel)
TimPosPresO ffP	REAL			True	False	Publish Only	Rider roll photocell OFF timeout (panel)
TimPosPresO nP	REAL			True	False	Publish Only	Rider roll photocell ON timeout (panel)
AllarmeOn	BOOL			False	False	Publish Only	Alarm active
BuzzerOn	BOOL			False	False	Publish Only	Panel buzzer command

Msg	INT			False	False	Publish Only	Message
Indice	INT			False	False	Publish Only	Alarm index
MotStop	BOOL			False	False	Publish Only	Motor stopped
MotOnH	BOOL			False	False	Publish Only	Motor running - normal speed
MotOnL	BOOL			False	False	Publish Only	Motor running - reduced speed
Clock1s	BOOL			False	False	Publish Only	1 s clock
ManMode	BOOL			True	False	Publish Only	Manual mode
SetVelMan	INT			True	False	Publish Only	% speed in manual
H_RsetCnt	BOOL			False	False	Publish Only	Counter reset from panel
I_RipaOpen	BOOL		BuiltInIO://cpu/#0/Input_Bit_09	False	False	Publish Only	Front guard open
O_OkScarIn	BOOL		BuiltInIO://cpu/#0/Output_Bit_07	False	False	Publish Only	Unloaders enable - closing
VelSetMan	REAL			True	False	Publish Only	Speed in manual
O_AccDec2	BOOL		BuiltInIO://cpu/#0/Output_Bit_05	False	False	Publish Only	Second acc/dec ramp
Aumenta	REAL			True	False	Publish Only	Acceleration coefficient in run
Diminuisci	REAL			True	False	Publish Only	Deceleration coefficient in run
DiminuisciH	REAL			True	False	Publish Only	Acceleration coefficient at start (orig. comment; name suggests decel)
AumentaH	REAL			True	False	Publish Only	Deceleration coefficient at start (orig. comment; name suggests accel)
O_SegnAcustico	BOOL		BuiltInIO://cpu/#0/Output_Bit_08	False	False	Publish Only	Horn / acoustic signal
Pot2_Fin	INT		BuiltInIO://opt/#0/Ch2_Analog_Input_Value	False	False	Publish Only	Potentiometer 2 value - Final speed -
I_StopMetri	BOOL		BuiltInIO://cpu/#0/Input_Bit_10	False	False	Publish Only	Stop on metre counter
I_RallEnd	BOOL		BuiltInIO://cpu/#0/Input_Bit_11	False	False	Publish Only	Slow down at end of roll
I_RotturaOn	BOOL		BuiltInIO://cpu/#0/Input_Bit_12	False	False	Publish Only	Paper break monitoring ON
I_ThAspira	BOOL		IOBus://unit#3/Ch1 Reset/External Input Status/Ch1 External Input 0 Status	False	False	Publish Only	Extractor fan thermal OK
I_FtRall	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN05 (Pin 1.3) _6000_05	False	False	Publish Only	Slow-down photocell

I_FtRotto	BOOL		ECAT://node #[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN07 (Pin 1.4) _6000_07	False	False	Publish Only	Paper break photocell
MtTot	REAL			True	False	Publish Only	Total metres
MtRal	REAL			True	False	Publish Only	Slow-down metres
VelVisR	REAL			True	False	Publish Only	Slow-down speed (displayed)
ImpEnc	INT			True	False	Publish Only	Encoder pulses
SetVelManR	INT			False	False	Publish Only	% slow-down speed in manual
RicRsetCnt	BOOL			False	False	Publish Only	Counter reset requested
Psw	INT			True	False	Publish Only	Password
Lingua	DINT			True	False	Publish Only	Language
I_RsetAlarm	BOOL		ECAT://node #[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN08 (Pin 2.4) _6000_08	False	False	Publish Only	Alarm reset
Fix10Vdc_1	UINT			False	False	Publish Only	Fixed 10 Vdc generator
DiaRuota	REAL			True	False	Publish Only	Encoder wheel diameter
AlarmCodeSV	ARRAY[0..32] OF BOOL			False	False	Publish Only	Panel alarms
ProtEscH	BOOL			False	False	Publish Only	Guards bypassed
RipaCloseH	BOOL			False	False	Publish Only	Front guard closed
Anal3	UINT			False	False	Publish Only	(?) Analog 3
Anal4	UINT			False	False	Publish Only	(?) Analog 4
Misurato	REAL			True	False	Publish Only	Measured distance
DiamAlbero	REAL			True	False	Publish Only	(?) Shaft diameter
MemNocciolo	BOOL			True	False	Publish Only	(?) Stored core value
MisuraNocciolo	REAL			True	False	Publish Only	(?) Core measurement
AI_Misura	INT		IOBus://unit#2/Ch1 Analog Input Value	False	False	Publish Only	Analog value - parent roll measurement
MisuraOn	BOOL			True	False	Publish Only	(?) Measurement enabled
DiamBobMis	REAL			True	False	Publish Only	(?) Measured roll diameter
DiamVasvaOff	REAL			True	False	Publish Only	(?) Diameter for tub/cradle off
DiamRall	REAL			True	False	Publish Only	(?) Slow-down diameter
DiamStop	REAL			True	False	Publish Only	(?) Stop diameter
H_RsetAlarm	BOOL			False	False	Publish Only	(?) Alarm reset from panel
Psw22	BOOL			False	False	Publish Only	(?) Password 22
Psw44	BOOL			False	False	Publish Only	(?) Password 44
I_RotoloOk	BOOL			False	False	Publish Only	(?) Roll OK
I_FtVascaOff	BOOL			False	False	Publish Only	(?) Tub/cradle-off photocell
I_FtEndBob	BOOL			False	False	Publish Only	(?) End-of-roll photocell